

Explaining LLMs



Introduction



The Parrot Who Wanted to Be a Poet

Once there was a **parrot** who didn't want to just repeat words...He **wanted** to understand them and **write poetry**.



Training Phase



Every day, the **parrot** listened to **humans** talk
Jokes, stories, tweets, arguments - he heard it all.
He didn't **memorize**; he noticed patterns.
After **once upon** a, usually comes **time**.

ONCE
UPON
A...

time



Probabilities



He started **guessing** what comes next.

Not by ***thinking***, but by **probability**.

That's how LLMs predict text - one token at a time.

"sun (5%)", "rain (40%)", "coffee (20%)"



Tokens



Then he noticed: words are made of ***smaller pieces!***
He **learned** to speak in tokens -
tiny language units **models** understand.



The Brain Map



The **parrot** built a mental map of how words connect.

"King Queen"

"Paris France"

"Dog Bark"

That web of links is the **Transformer** at work.



Prompting and Generation



Soon everyone asked the **parrot** for **poems** and **stories**

He read the ***prompt***: "**Write a love poem.**"

It learns using And started creating - word by word, beautifully.

That's **inference**: **generating text** from a prompt.



Hallucination



Then someone asked:

- What's the capital of **Mars**?
- The **parrot** proudly said: "**Banana City!**"
- He **sounds** **smart**, but doesn't **know**.
- LLMs don't understand truth _ they **predict patterns**.



Fine-Tuning and Alignment



Scientists trained him to be **polite**, **helpful**, and **safe**.
"Be kind." "Avoid nonsense." "Respect humans."
That's **fine-tuning** teaching **models** to follow values.



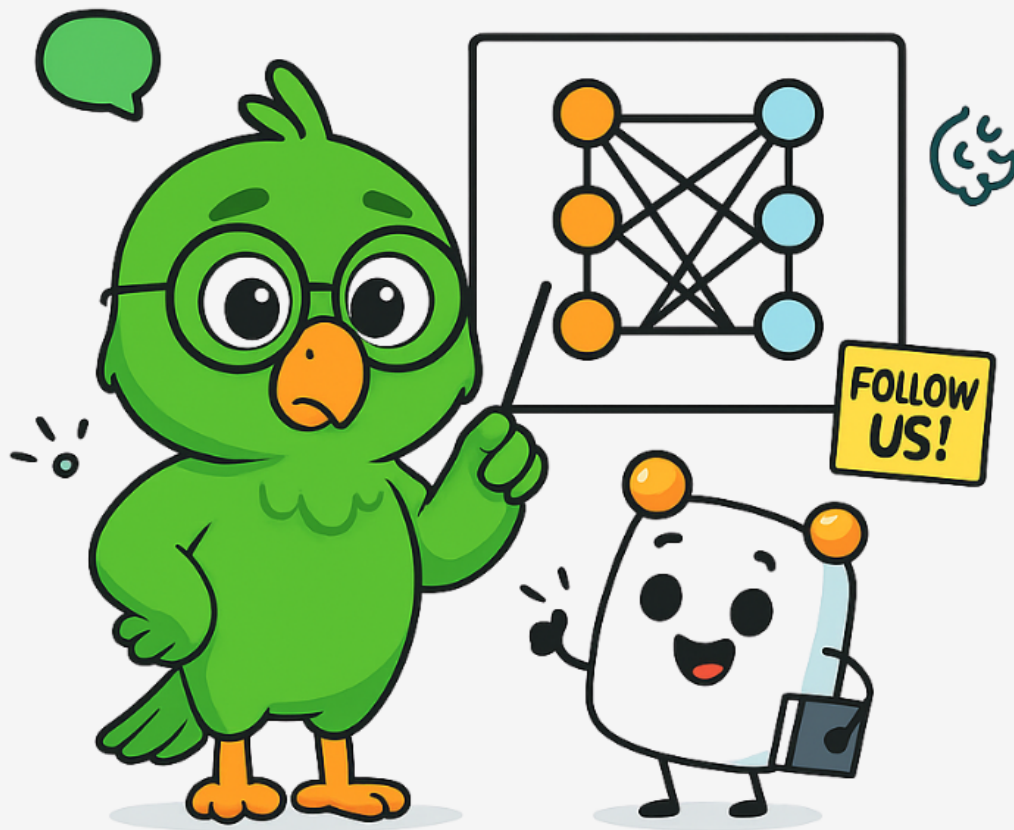
Wisdom



The **parrot** became a true poet.
He doesn't feel love...
but he **learned** how to express it **beautifully**.
That's what an **LLM** really is -
A **parrot** that mastered the ***art of prediction.***



THE LLMs ARE LEARNING...



You should too

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